1. Purpose and limitations

1 Purpose and limitations

1.1 Scope

This fictional manual exists to test the Noesis document ingestion and retrieval pipeline. The pump, model, and scenarios are invented for software testing.

1.1.1 Synthetic status

This is NOT an OEM manual. No real equipment specifications, measured results, or validated maintenance limits are provided. See Section 2, Safety, for the general safety context.

1.2 Use limitations

Never use this document to select, install, operate, repair, or maintain real equipment. Consult the actual manufacturer's manual and approved site procedures.

2. Safety

2 Safety

WARNING

This synthetic document is not an authorized safety procedure. Before maintenance, follow applicable site rules, equipment instructions, energy-isolation requirements, and verification steps. Only qualified and authorized personnel may perform assigned work.

2.1 General safety

Identify hazards before starting work. Do not bypass guards, interlocks, or approved procedures. See Section 2.1.1 for the energy-isolation topic.

2.1.1 Energy isolation

Follow the site's approved lockout/tagout process and verify the required safe state before work. This text is an overview, not a substitute for the applicable procedure.

3. Fictional equipment record

3 Fictional equipment record

3.1 Equipment identification

Equipment tag: P-101A
Alternate tag display: P 101 A
Another tag display: p101a
Description: Fictional centrifugal water-transfer pump
Fictional model: SYN-100
Inspection or test data: None collected

3.1.1 Tag interpretation

The three tag strings above are surface forms intended for normalization tests. A canonical tag must not accidentally merge P101 with P101A. API 610 is a standard designation, not an equipment tag in this fixture.

3.2 Basic components

A centrifugal pump may include an impeller, casing, shaft, bearings, shaft seal, suction connection, and discharge connection. Actual designs vary by manufacturer and model.

4. Maintenance and condition observations

4 Maintenance and condition observations

4.1 Bearing observations

Record visible condition, unusual noise, lubrication observations, and vibration relative to an established baseline. Consult the equipment-specific limits.

4.1.1 Vibration observations

A change in vibration may have several possible causes. Misalignment, imbalance, looseness, bearing damage, and hydraulic conditions are illustrative topics only.

4.2 Lubrication observations

Record the lubricant identity and actual observed condition when those data are available. Do not infer a missing measurement or assume that one lubricant fits every pump.

5. Troubleshooting

5 Troubleshooting

5.1 Low discharge flow

Possible investigation areas include liquid supply, suction restriction, air ingress, valve position, rotation, and operating conditions. See Section 2, Safety, before work.

5.2 Elevated vibration

Possible contributors include misalignment, imbalance, looseness, bearing damage, or hydraulic conditions. Compare measurements with valid equipment-specific baselines.

Table 5.1 · Illustrative troubleshooting guide

Observed condition	Possible area	Next step
Low discharge flow	Supply or suction	Check approved procedure
Elevated vibration	Alignment or bearing	Compare with baseline
Seal-area leakage	Seal or connections	Follow site procedure

6. Inspection records

6 Inspection records

6.1 Recording observations

Record the equipment tag, date, observation, measurement, units, and source of the measurement when available. Keep observations distinct from interpretations and recommendations.

6.1.1 Missing information

If a value is unavailable, mark it as unavailable. Never invent missing measurements, thresholds, or inspection results.

6.2 Illustrative inspection notes

Equipment: P-101A
Observed condition: Not supplied
Measured vibration: Not supplied
Assessment: Cannot be determined from this fixture

7. Document control

7 Document control

7.1 Document identity

Document ID: SYN-PUMP-MAN-001
Status: Fictional training document · NOT AN OEM MANUAL
Fictional model: SYN-100

7.1.1 Permitted use

This fixture tests PDF extraction, heading detection, table parsing, warning isolation, equipment-tag normalization, and source-grounded retrieval. It does not validate real maintenance procedures or equipment limits.

7.2 Cross-reference

Sections 1, 3, and 5 refer readers to Section 2, Safety. The reference is intentionally repeated to test section-aware chunking and retrieval.